

Tools of Budget Analysis

Danny Yagan
UC Berkeley
Spring 2026

Budget Analysis: Today's Big Questions

- ① Are deficits bad?
- ② Is the U.S. in trouble?

- **Debt:** The amount borrowed by government through bonds to individuals, firms, or foreign governments. Debt is a **stock**
- **Deficit:** government's spending + interest payments on debt minus government revenues in a given year. A negative deficit is called a surplus. Deficit is a **flow**
- Evolution of debt from year to year:

$$\text{Debt}_{t+1} = \text{Debt}_t + \text{Deficit}_t = \text{Debt}_t \cdot (1 + r_t) + \text{Spending}_t - \text{Revenue}_t$$

with r_t interest paid on government debt

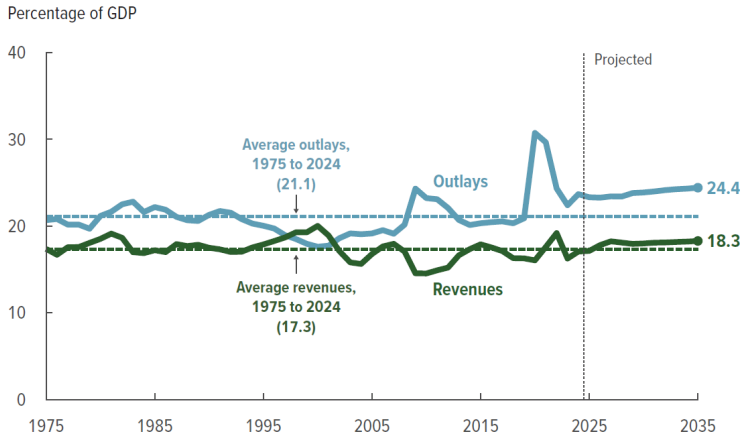
- Primary Deficit = Spending - Revenue

Government Budgeting: U.S. Example

- 2025Q3: \$30Tr, equal to just under 100% (97%) of GDP
- Was about one third of GDP in the mid-2000s, then three-quarters of GDP in the mid-2010s
- US government owns assets worth about 80% of GDP

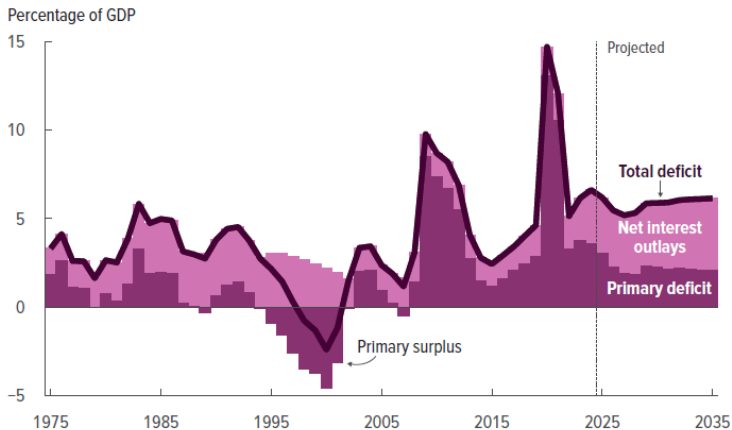
Total Outlays and Revenues

Measured as a percentage of GDP, federal outlays in CBO's projections exceed their 50-year average every year from 2025 to 2035. Revenues remain below their 50-year average in 2025 but rise above it thereafter.



Total Deficit, Net Interest Outlays, and Primary Deficit

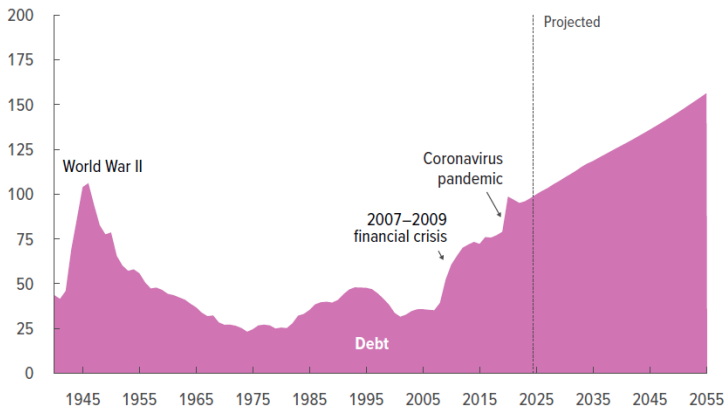
In CBO's projections, the total deficit—the amount by which outlays exceed revenues—equals 6.1 percent of GDP in 2035. By that year, net interest payments grow to 4.1 percent of GDP and account for about one-sixth of all federal spending. The primary deficit (which excludes those payments) equals 2.1 percent of GDP in 2035.



Federal Debt Held by the Public

Debt increases in relation to GDP, exceeding any previously recorded level in 2029 and continuing to soar through 2055. It is on track to increase even more thereafter.

Percentage of GDP



What Is Government Debt?

- Government debt increases private wealth at the expense of public wealth (but no direct effect on national wealth = private + public wealth in closed economy)
- Government debt is not borrowing on the back of future generations but rather changing the distribution of future wealth
- Sustainable government debt can be a free lunch for the first generation. E.g. permanently transfer consumption from young to old, so first old generation consumes more while all others consume same
- Unsustainable government debt commits future governments to higher taxes or lower spending, or a version of default

Government Debt Sustainability

- $\text{Debt}_t = \text{Debt}_{t-1} + \text{Deficit}_t$
- With r_t nominal interest on debt:

$$\text{Debt}_t - \text{Debt}_{t-1} = r_t \cdot \text{Debt}_{t-1} + \text{PrimaryDeficit}_t$$

- Debt as share of GDP can snowball when r_t exceeds nominal GDP growth rate g_t :

$$\frac{\text{Debt}_t}{\text{GDP}_t} - \frac{\text{Debt}_{t-1}}{\text{GDP}_{t-1}} = \frac{r_t \cdot \text{Debt}_{t-1}}{\text{GDP}_t} - \frac{g_t \cdot \text{Debt}_{t-1}}{\text{GDP}_t} + \text{PrimaryDeficit}_t$$

- US can run a permanent primary deficit if r_t is permanently less than g_t :

$$\text{PrimaryDeficit}_t = \frac{g_t - r_t}{1 + g_t} \cdot \frac{\text{Debt}}{\text{GDP}}$$

Future generations have to pay for a current generation's primary deficits.

- A. True
- B. False
- C. Uncertain



Sources: OMB; St. Louis Fed

myf.red/g/lKfK
[Customize](#) | [Download Data](#)

Historical Experiences with Government Debts

Countries have incurred debts of 100-300% of GDP due to crises, wars, foreign coercion. Several ways out:

- ① Repay debt with primary surpluses over many decades (UK in 19th century): wealthy debt holders win, govt spending suffers
- ② Repudiate debt (Soviet Union with Russian debt in 1917): foreign wealthy debt holders lose but hard to borrow afterwards
- ③ Inflate debt away (US, EU countries after World War II): Debt holders (and many others) lose, risk of hyperinflation if govt cannot fund itself with taxes (Germany 1920s)
- ④ Exceptional wealth taxes: (partly Germany, Japan after World War II): wealthy in general lose not just debt holders

Debt financing looks attractive in short-run. Who ultimately pays is more opaque than with tax financing.

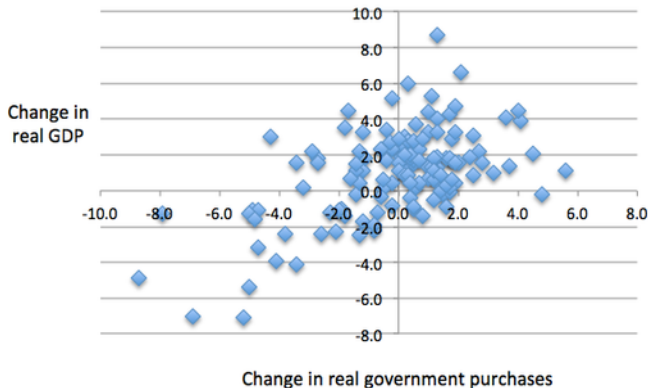
The U.S. Federal Process

- Taxes, spending, and debt ceiling are decided by Congress and the President
- New laws requires majority vote both in House and in Senate along with President's signature (veto power). Senate usually requires 60/100 super-majority (filibuster).
- Two forms of spending:
 - Entitlement spending: Mandatory funds for programs for which funding levels are automatically set by the number of eligible recipients (ex: medicare, social security)
 - Discretionary spending: Optional spending set by appropriation levels each year, at Congress's discretion (ex: defense)
- Failure to pass appropriation results in Federal government shutdown (right now!)

Short-Run Effects of the Government on the Macroeconomy

- Keynesian theory (IS-LM macro model): More government spending or tax cuts stimulates the economy in the short-run [and conversely]
- Short-run stabilization: Govt can use taxes and spending policies to smooth the peaks and troughs of the business cycle
 - Automatic stabilization: Policies that automatically alter taxes or spending in response to economic fluctuations to offset changes in household consumption levels (ex: unemployment insurance, progressive taxation, corporate profits tax)
 - Discretionary stabilization: Policy actions taken by the government in response to business cycle (ex: Fiscal stimulus with Spring 2008 rebate checks, 2009-12 Obama stimulus, unemployment insurance extensions)

Government spending and growth, 2010-2013



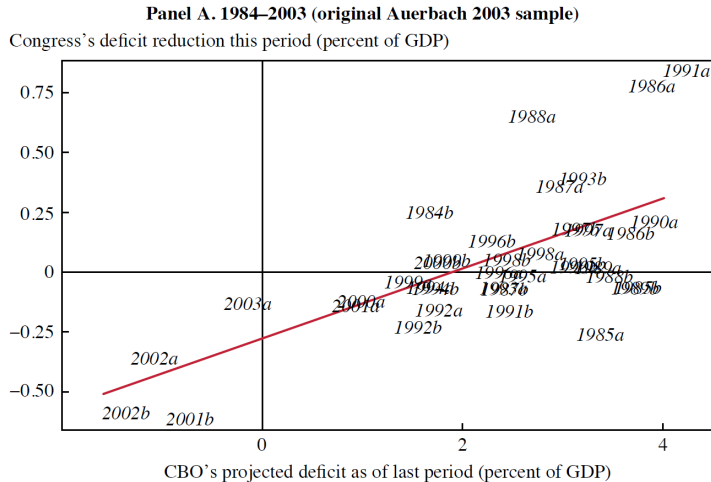
% changes in annual real govt spending and changes in real GDP, 33 EU countries, 2010-11, 2011-2, 2012-3 (=99 dots). Source: Krugman NYtimes blog, January 6, 2015

Does Government Do What It Should?

- Most economists: Run high deficits in recessions, and run low enough deficits (or run surpluses) otherwise to keep debt/GDP sustainable
- How low of a deficit? Depends on debt/GDP, interest rate, growth rate
- Simple “fiscal feedback” rule: Cut the deficit when projected deficit is high or when output gap is high

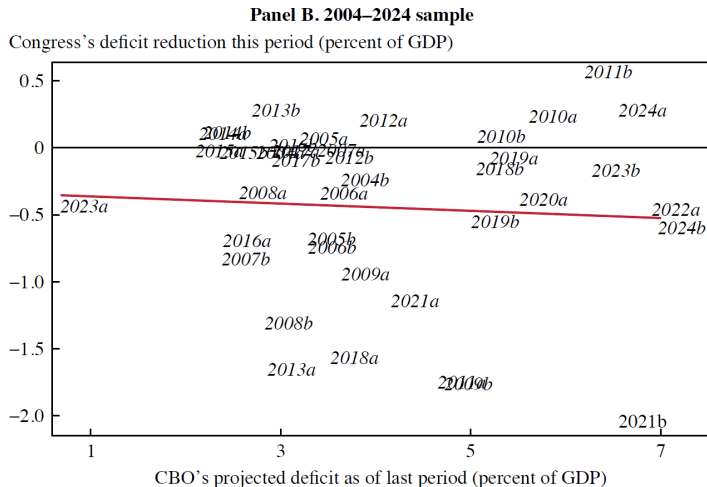
$$DEFICITREDUCTION_t = \alpha + \beta \mathbb{E} [DEFICIT_{t-1}] + \gamma OUTPUTGAP_{t-1} + \epsilon_t$$

1980s-90s: Congress cut the deficit when it was high



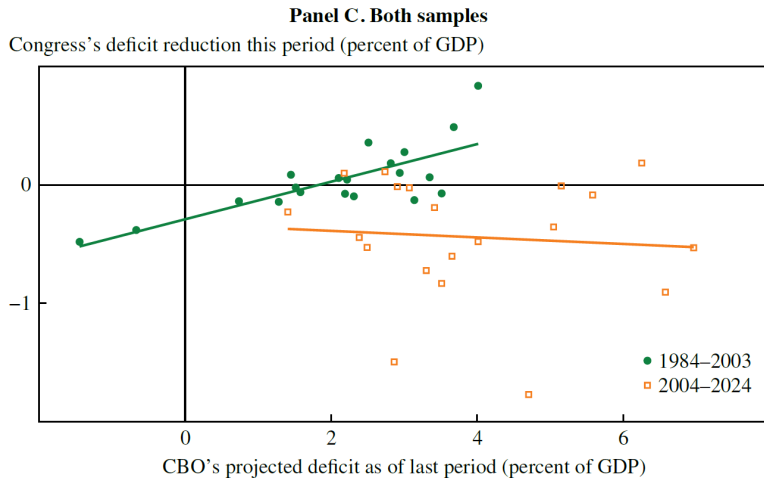
Source: Auerbach and Yagan (2024)

Last two decades: Congress stopped fiscal feedback



Source: Auerbach and Yagan (2024)

Stark change in Congressional behavior



Source: Auerbach and Yagan (2024)

Does it matter who is in charge?

- Recall that power is split among the President, the House, and the Senate
- Does who is in charge matter for fiscal discipline?
- Let's go to the data!

- Deficit should be high in recessions and low otherwise
- Deficits are now high even in good times
- Chicken-and-egg problem: High interest rates could scare Congress into cutting deficits, but then interest rates don't need to be high

References

- Auerbach, A. J. and Yagan, Danny. “Robust Fiscal Stabilization”, Brookings Papers on Economic Activity 2024, fall, pp. 239-293. [\[web link\]](#)
- Congressional Budget Office “The Budget and Economic Outlook: Fiscal Years 2025 to 2035” [\[web link\]](#)
- Gruber, Jonathan. Public Finance and Public Policy, 2016 Worth Publishers, Chapter 4
- Krugman, Paul. “The Record of Austerity”, New York Times Blog, January 6, 2015. [\[web link\]](#)
- Yagan, Danny. “The Growth Dividend and Excess Interest”, Tax Policy and the Economy 2025, volume 39(1), pp. 29-52. [\[web link\]](#)